Chapter 16. A Taste of Topology

Compactness and connectedness of matrix groups

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Identify $\mathfrak{M}_{n,n}(\mathbb{R}) = \mathbb{R}^{n^2}$ and $\mathfrak{M}_{n,n}(\mathbb{C}) = \mathbb{C}^{n^2} = \mathbb{R}^{2n^2}$ (coordinates in any fixed order), with the Euclidean norm, distance and continuity.

16.1 Matrix group isomorphisms

Motivation. Matrix groups live in Euclidean space, so they have shapes as well as group structure.

Exercise 16.1.1

Under $\mathbb{C}^n = \mathbb{R}^{2n}$, the Hermitian dot product of $\mathbb{C}^n$ and the dot product of $\mathbb{R}^{2n}$ give the same norm (and metric).

Solution. $|a+ib|^2 = a^2 + b^2$ coordinatewise (compare Remark 10.2.20). □

Example 16.1.2

$\det : \mathfrak{M}_{n,n}(\mathbb{R}) \to \mathbb{R}$ is a polynomial map, hence continuous, so $\mathrm{GL}_n(\mathbb{R}) = \det^{-1}(\mathbb{R} - \{0\})$ is an open subset of $\mathbb{R}^{n^2}$; likewise $\mathrm{GL}_n(\mathbb{C})$ is open in $\mathbb{R}^{2n^2}$.

Example 16.1.3

$\mathrm{SL}_n(\mathbb{R}) = \det^{-1}(1)$ is a closed subset of $\mathbb{R}^{n^2}$; it is a **closed subgroup** of $\mathrm{GL}_n(\mathbb{R})$. Likewise $\mathrm{SL}_n(\mathbb{C})$ is a closed subgroup of $\mathrm{GL}_n(\mathbb{C})$.

Example 16.1.4

$\mathbf{O}(n) = \{A : A^tA - I = 0\}$ is the common zero set of n^2 polynomials in n^2 variables, hence closed; it is a closed subgroup of $\mathrm{GL}_n(\mathbb{R})$ (and a closed subset of $\mathfrak{M}_{n,n}(\mathbb{R})$).

Exercise 16.1.5

$\mathbf{U}(n)$ is a closed subgroup of $\mathrm{GL}_n(\mathbb{C})$.

Solution. $A^*A = I$ is a system of polynomial equations in the real and imaginary parts of the entries. □

Exercise 16.1.6

(a) $\mathbf{SO}(n)$ is a closed subgroup of $\mathbf{O}(n)$.
(b) $\mathbf{SU}(n)$ is a closed subgroup of $\mathbf{U}(n)$.

Solution. Each is cut out by an additional polynomial equation $\det A = 1$ (real and imaginary parts). □

Question 16.1.7

For $F = \mathbb{R}$ in Exercise 12.4.12, is $\operatorname{im}\psi$ a closed subgroup of $\mathrm{SL}_3(\mathbb{R})$?

Definition 16.1.8 ("Same shape")

For $X \subseteq \mathbb{R}^r$, $Y \subseteq \mathbb{R}^s$, a bijection $f : X \to Y$ with f and f^{-1} continuous is a **homeomorphism** ("same-shape map"); then X and Y have the **same shape** (are homeomorphic).

Exercise 16.1.9

"Same shape" is an equivalence relation.

Solution. Identities, inverses and composites of homeomorphisms are homeomorphisms. □

Exercise 16.1.10

If $f : X \to Y$ is a homeomorphism and $Z \subseteq X$: (a) Z open in $X \Rightarrow f(Z)$ open in Y; (b) Z closed in $X \Rightarrow f(Z)$ closed in Y.

Solution. $f(Z) = (f^{-1})^{-1}(Z)$, and preimages of (relatively) open/closed sets under the continuous map f^{-1} are open/closed. □

Example 16.1.11

$(a, b) \subseteq \mathbb{R}$ and $X = \{(x, 1)^t : a < x < b\} \subseteq \mathbb{R}^2$ have the same shape, although (a, b) is open in $\mathbb{R}$ and X is not open in $\mathbb{R}^2$. Any two open intervals have the same shape.

Example 16.1.12

$\exp : \mathbb{R} \to (0, \infty)$ is a homeomorphism (Example 11.5.7).

Example 16.1.13

$\tan : (-\frac{\pi}{2}, \frac{\pi}{2}) \to \mathbb{R}$ is a homeomorphism.

Exercise 16.1.14

$[a, b)$ and $[a, \infty)$ have the same shape.

Solution. $x \mapsto a + \tan\left(\frac{\pi}{2} \cdot \frac{x-a}{b-a}\right)$ is a continuous bijection $[a, b) \to [a, \infty)$ with continuous inverse. □

So openness, closedness and boundedness are not shape invariants.

Exercise 16.1.15

A circle and an ellipse have the same shape. What about a circle and a square?

Solution. $(x, y) \mapsto (ax, by)$ maps the unit circle onto $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$. The square (its boundary) also has the same shape as the circle, via radial projection $v \mapsto v/\|v\|$ from the centre. □

Exercise 16.1.16

Explain how a topologist turns a soccer ball with a needle hole inside out.

Solution. A sphere with a small hole has the shape of a disk: stretch the hole, flip the disk, shrink the hole back. □

Exercise 16.1.17

(a) Do $\mathbb{R}$ and $\mathbb{R}^2$ have the same shape?
(b) $\mathbb{R}$ and S^1?

Solution. [Supplement] (a) No: removing one point disconnects $\mathbb{R}$ but not $\mathbb{R}^2$, and a homeomorphism restricts to a homeomorphism of the complements.

(b) No: S^1 is compact and $\mathbb{R}$ is not (compactness is preserved, Proposition 16.2.1). □

Exercise 16.1.18

For a matrix group $G \leq \mathrm{GL}_n(F)$: (a) left multiplication $\lambda_g : G \to G$ is a homeomorphism (inverse $\lambda_{g^{-1}}$); likewise right multiplication; (b) inversion $x \mapsto x^{-1}$ is a homeomorphism of G; (c) for fixed g, $\mathrm{Int}(g) : G \to G$ is a homeomorphism.

Solution. Matrix multiplication is polynomial in the entries, hence continuous. Inversion is continuous by the adjugate formula $A^{-1} = \frac{1}{\det A} \operatorname{adj} A$ and is its own inverse. $\mathrm{Int}(g) = \lambda_g \circ \rho_{g^{-1}}$. □

Observation 16.1.19

If H is a subgroup of a matrix group G and $g \in G$, then H and the coset gH have the same shape.

Proof. $\lambda_g(H) = gH$, and λ_g restricts to a homeomorphism $H \to gH$ (Exercise 16.1.18(a)). ■

Definition 16.1.20 (Matrix group isomorphism)

For matrix groups $G \leq \mathrm{GL}_n(F) \subseteq \mathbb{R}^N$ and $H \leq \mathrm{GL}_m(F') \subseteq \mathbb{R}^M$ ($F, F' \in \{\mathbb{R}, \mathbb{C}\}$, possibly different), $\varphi : G \to H$ is a **matrix group isomorphism** if it is a group isomorphism and a homeomorphism; then G and H are **isomorphic as matrix groups**, $G \approx_{\mathrm{mat}} H$.

Example 16.1.21

$\exp : (\mathbb{R}, +) \to (\mathbb{R}^+, \times)$ is a matrix group isomorphism.

Exercise 16.1.22

$\mathbf{SO}(2) \approx_{\mathrm{mat}} S^1 \approx_{\mathrm{mat}} \mathbf{U}(1)$.

Solution. $R_\theta \mapsto e^{i\theta}$, i.e. $\left(\begin{smallmatrix} x & -y \\ y & x \end{smallmatrix}\right) \mapsto x + iy$, is a group isomorphism (Exercise 11.9.20) and restricts a linear map with linear inverse; $S^1 = \mathbf{U}(1)$. □

Observation 16.1.23

If $H \leq G \leq \mathrm{GL}_n(F)$ and $g \in G$, then $H \approx_{\mathrm{mat}} gHg^{-1}$.

Proof. $\mathrm{Int}(g)$ restricts to a group isomorphism $H \to gHg^{-1}$, which is a homeomorphism by Exercise 16.1.18(c). ■

16.2 Compactness and connectedness

Motivation. $\mathbf{O}(n)$ and $\mathbf{U}(n)$ are compact, and the spectral theorems make $\mathbf{SO}(n)$, $\mathbf{U}(n)$ connected.

In $\mathbb{R}^N$, X is **compact** iff it is closed and bounded, and (for the sets considered here) **connected** means arcwise connected.

Proposition 16.2.1

If $f : X \to Y$ is continuous ($X \subseteq \mathbb{R}^N$, $Y \subseteq \mathbb{R}^M$): (a) X compact $\Rightarrow$ $f(X)$ compact; (b) X connected $\Rightarrow$ $f(X)$ connected.

Proof. Well known from analysis. ■

So compactness and connectedness are shape invariants.

Example 16.2.2

$\mathrm{GL}_n(\mathbb{R})$ and $\mathrm{SL}_n(\mathbb{R})$ ($n \geq 2$) are not bounded in $\mathbb{R}^{n^2}$ (e.g. $\mathrm{diag}(t, t^{-1}, 1, \dots)$), hence not compact; likewise $\mathrm{GL}_n(\mathbb{C})$, $\mathrm{SL}_n(\mathbb{C})$.

Example 16.2.3

For $A \in \mathbf{O}(n) \subseteq \mathbb{R}^{n^2}$, $\|A\| = \sqrt{n}$ (its columns are unit vectors, Observation 9.3.6(5)), so $\mathbf{O}(n)$ is bounded; it is closed (Example 16.1.4), hence a compact group.

Example 16.2.4

$\mathbf{SO}(n)$, a closed subgroup of $\mathbf{O}(n)$, is compact.

Exercise 16.2.5

$\mathbf{U}(n)$ and $\mathbf{SU}(n)$ are compact.

Solution. As in Examples 16.2.3, 16.2.4: $\|A\| = \sqrt{n}$ for $A \in \mathbf{U}(n)$ (Exercise 10.6.9(5), Exercise 16.1.1), and they are closed (Exercises 16.1.5, 16.1.6). □

Example 16.2.6

(a) $\mathbf{SO}^\circ(1,1)$ is unbounded in $\mathbb{R}^4$ ($\cosh t \to \infty$), so $\mathbf{SO}^\circ(1,1)$ and $\mathbf{O}(1,1)$ are not compact. (b) Alternatively, $\mathbf{SO}^\circ(1,1) \approx_{\mathrm{mat}} (\mathbb{R}, +)$ (Exercise 13.4.5; the inverse $\left(\begin{smallmatrix} x & y \\ y & x \end{smallmatrix}\right) \mapsto \operatorname{arsinh} y$ is continuous), and $\mathbb{R}$ is not compact. (c) So $\mathbf{SO}(2) \approx_{\mathrm{mat}} S^1$ (compact) and $\mathbf{SO}^\circ(1,1)$ are not isomorphic as matrix groups, nor even as groups: $\mathbf{SO}(2)$ has elements of every finite order, $\mathbf{SO}^\circ(1,1) \approx \mathbb{R}$ none (cf. Exercise 13.4.6).

Example 16.2.7

$\det : \mathrm{GL}_n(\mathbb{R}) \to \mathbb{R}^\times = \mathbb{R} - \{0\}$ is a continuous surjection and $\mathbb{R}^\times$ is not connected; so $\mathrm{GL}_n(\mathbb{R})$ is not connected.

Exercise 16.2.8

$\mathbf{O}(n)$ and $\mathbf{O}(1,1)$ are not connected.

Solution. det maps each continuously onto $\{\pm 1\}$, which is not connected. □

Proposition 16.2.9

For $X_i \subseteq \mathbb{R}^{N_i}$, identify $X_1 \times \cdots \times X_r \subseteq \mathbb{R}^{N_1+\cdots+N_r}$. (a) If all X_i are compact, so is the product.
(b) If all X_i are connected, so is the product.

Proof. Well known. ■

Definition 16.2.10 (Torus)

$\mathbf{T} \subseteq \mathbb{R}^N$ is a **torus** if it has the same shape as $S^1 \times \cdots \times S^1$. A torus is compact and connected; $S^1 \times S^1$ has the shape of a doughnut.

Exercise 16.2.11

$\mathbf{T}_{\mathbf{U}(n)}$, $\mathbf{T}_{\mathbf{SU}(n)}$ (Definition 15.2.7) and $\mathbf{T}_{\mathbf{SO}(n)}$ (Definition 15.4.4) are tori.

Solution. The isomorphisms of Exercises 15.2.8, 15.4.5 are coordinate maps with continuous inverses. □

Corollary 16.2.12 (Corollary of the spectral theorems)

$\mathbf{U}(n)$, $\mathbf{SU}(n)$ and $\mathbf{SO}(n)$ are connected groups.

Proof. For $\mathbf{SO}(n)$ (the others likewise via Corollaries 15.2.9, 15.2.11): $\mathbf{SO}(n) = \bigcup_U U\,\mathbf{T}_{\mathbf{SO}(n)}\,U^{-1}$ (Corollary 15.4.6), a union of connected sets (Observation 16.1.23, Exercise 16.2.11) all containing I. ■

Every $X \subseteq \mathbb{R}^N$ is uniquely the disjoint union of its **connected components** (maximal connected subsets), its **component decomposition**.

Observation 16.2.13

(a) $\mathbf{SO}(n)$ is a connected component of $\mathbf{O}(n)$.
(b) The coset decomposition $\mathbf{O}(n) = \mathbf{SO}(n) \amalg (\mathbf{O}(n) - \mathbf{SO}(n))$ is the component decomposition of $\mathbf{O}(n)$.

Proof. $\mathbf{SO}(n)$ is connected (Corollary 16.2.12), and so is $\mathbf{O}(n) - \mathbf{SO}(n) = S \cdot \mathbf{SO}(n)$ (Observation 16.1.19). No connected subset can meet both, since det would map it continuously onto $\{\pm 1\}$. ■

Observation 16.2.14

(a) $\mathbf{SO}^\circ(1,1)$ is a connected component of $\mathbf{O}(1,1)$.
(b) The coset decomposition of $\mathbf{O}(1,1)$ modulo $\mathbf{SO}^\circ(1,1)$ (§13.4) is its component decomposition.

Proof [Supplement]. $\mathbf{SO}^\circ(1,1) \approx \mathbb{R}$ and its cosets are connected (Observation 16.1.19), and the continuous det and $\operatorname{sign}(A_{11})$ ($|A_{11}| \geq 1$) separate the four cosets. ■

(The identity component is always a subgroup; proof postponed.) So $\mathbf{O}(2)$ is two circles and $\mathbf{O}(1,1)$ four lines.